Aggregate Authentication under Proxy Re-encryption: A Security Boundary between Ciphertext- and Class-Granular Confidentiality

Anonymous manuscript for review

Abstract

Encrypted collection systems may need to compress authentication before a proxy delegates decryption to a new recipient. A direct composition is difficult: aggregate authentication normally binds the ciphertext transcript, whereas proxy re-encryption changes that transcript. We study a commit–encrypt– aggregate–delegate comparator in which each sender signs a commitment to its message, Schnorr responses are half-aggregated, and the proxy changes only the encryption point. Aggregation and delegation then commute exactly. This identity follows from the invariant signed statement and supplies the functional baseline for the security analysis.

The same design exposes a confidentiality boundary. Public aggregate verification does not validate the encryption point. A corrupt delegatee can shift the challenge point, obtain one permitted transformation, remove the known shift, and recover the challenge opening. The attack wins a natural ciphertext-granular game with advantage one and uses neither a hash collision nor a decryption-oracle query. We therefore separate ciphertext granularity from a weaker class-granular game that refuses every target-class transformation to a corrupt recipient. The latter notion is strictly weaker than honest- reencryption-attack security.

For the class-granular game, we give a candidate random-oracle reduction under a gap variant of Divisible Computational Diffie–Hellman. Its central simulation samples the scalar proxy token directly after a uniform-mask hybrid, which eliminates the algebraic extraction premise required by an earlier proof route. The bound accounts explicitly for honest delegatees, proof-hash prequeries, commitment queries, and decryption attempts. Independent class keys give an exact fixed-target scope composition. An executable artifact fixes the games, encodings, attack, and hybrid bookkeeping; its 286 unique tests and 33 deterministic trace replays pass without failures or skips. **Keywords:** proxy re-encryption; aggregate signatures; signcryption; chosen-ciphertext security; random-oracle model.

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Proof status. The theorem statements in Sections 7 and 8 are candidate results until a cryptographer independently checks every oracle simulation and loss term. The negative result in Section 6 and the exact identities in Sections 4 and 6 are algebraic and are covered by executable tests.

1 Introduction

Data collection systems often need three operations at once. Many senders produce encrypted records, an intermediary compresses their authentication, and a proxy later delegates decryption to another recipient without learning the messages. A direct combination is fragile. Aggregate authenticators tend to bind the ciphertext transcript, while proxy re-encryption changes that transcript. Removing the changing term from the authenticator can restore commutation but can also create a receiver-insider forgery.

The syntactic conflict can be resolved by committing to the plaintext, encrypt it, sign the commitment rather than the ciphertext, and aggregate the signatures. The signed statement is then invariant under delegation. This idea combines signcryption and joint signature–encryption disciplines [Zhe97, ADR02], single-hop proxy re-encryption constructions and security models [BBS98, AFGH06, CH07], and incremental half-aggregation of Schnorr signatures [CGKN21, CZ22]. We call the resulting comparator CEA. The name is descriptive: it denotes commit, encrypt, and authenticate.

Exact commutation does not settle confidentiality. In CEA, public verification authenticates the message commitment, not the encryption point. This distinction is harmless for an origin recipient, who checks the commitment after decryption. It is harmful for a proxy that must decide whether a submitted ciphertext may be transformed. A shifted encryption point retains a valid aggregate authenticator, and the shift survives proxy transformation in a form that a corrupted delegatee can remove. Thus a construction may have correct delegation, authentic origin statements, and exact aggregate commutation while failing a natural per-ciphertext proxy security game.

This paper separates the two confidentiality levels. The stronger notion, G-CONF-CT, permits the honest proxy to serve nonderivative target-class ciphertexts to a corrupted delegatee. We show that CEA fails this notion with advantage one. The weaker notion, G-CONF-CLASS, treats the independently keyed class—an origin, tag, and epoch—as the authorization boundary. It never serves that target class to a corrupted party. The distinction matches the construction: a proxy–delegatee coalition can recover the class key, making the class itself the authorization unit.

1.1 Contributions

Our first contribution is a precise negative result. We give an algebraic A-mauling attack against G-CONF-CT. The attack changes only an encryption point, passes public aggregate verification, and uses one permitted proxy query to recover the challenge opening. The attack is purely algebraic and applies to the specified construction independently of implementation details.

Second, we freeze a class-granular game and give a candidate quantitative reduction. The main technical correction is type-level but consequential. The proxy secret $\Omega = \omega k$ is a scalar, not the point ωK. After a random-oracle hybrid makes the token mask uniform, a simulator can sample Ω directly and answer every honest-delegatee transform exactly as $\bar{A} = \Omega^{-1}A$, using only the sampled proxy scalar. A separate gap-DCDH reduction accounts for exposure of the masked token point.

Third, we show that independently keyed non-target classes compose exactly with the class theorem. For a target fixed before setup, the resulting G-SCOPE game preserves the same assumption and bound. Adaptive target selection, in-class proxy–delegatee collusion, revocation, and concrete-hash analysis form separate security questions.

Finally, we supply a reproducible artifact. It includes the frozen game interfaces, canonical encodings, the attack, hybrid probes, and a complete sender-to-delegatee execution. Tests establish functional and algebraic claims; the computational claims rest on the accompanying reductions.

Taken together, these results identify the security boundary of invariant commitment authentication under proxy transformation: ciphertext-granular confidentiality fails by a concrete algebraic attack, whereas class-granular confidentiality admits an explicit candidate bound. This separation, rather than the commutation identity alone, is the paper's central contribution.

2 Related work and positioning

Proxy re-encryption and the security hierarchy. Early PRE schemes established bidirectional and unidirectional delegation mechanisms [BBS98, AFGH06]; Canetti and Hohenberger then formalized chosen-ciphertext-secure PRE, including derivative restrictions and key registration issues [CH07]. Replayable CCA security masks challenge-equal outputs rather than rejecting every related ciphertext, a pattern used by our decryption rule [CKN03]. More recent work argues that CPA is too weak for PRE and introduces honest-reencryption-attack (HRA) security, which permits honest reencryption from an honest delegator to a corrupt recipient [Coh19]. Tight and adaptively secure HRA constructions now exist in the standard model [CCG$^+$25, LCB$^+$25]. This hierarchy is essential for interpreting our result: G-CONF-CT permits proxy service on adversarially modified, nonderivative target ciphertexts to a corrupt delegatee and is therefore CCA-like and stronger than HRA in this respect; G-CONF-CLASS refuses all target-class service to corrupt recipients and is intentionally weaker than HRA. This placement makes CEA a comparator for the granularity boundary studied here.

Joint authentication and encryption. Signcryption originates with Zheng [Zhe97]. An, Dodis, and Rabin show that natural signature–encryption compositions require careful joint security definitions [ADR02]. Fuchsbauer, Plouviez, and Seurin prove Schnorr-signed ElGamal CCA security in the algebraic group and random-oracle models [FPS20]. Their public proof of encryption randomness motivates a repair for our mauling attack, but a compressed multi-record repair would still need per-record knowledge soundness. Signatures on randomizable ciphertexts instead authenticate a plaintext-equivalence class across rerandomization [BF20]; their transformation scope is rerandomization rather than recipient-changing PRE.

Aggregate authentication. Pairing-based aggregate signatures achieve compact aggregation [BGLS03]. For Schnorr-type signatures, noninteractive half aggregation compresses the response scalars while retaining the commitment points [CGKN21]. Chen and Zhao give a tight AGM–ROM analysis and an incremental construction [CZ22]. Recent alternatives occupy different points in the design space: DahLIAS gives two-round, constant-size, pairing-free aggregation under an algebraic one-more-DL assumption [NRS26]; transcript rebinding amortizes interactive transcripts across signing domains [ZWL26]; and plain-model constant-size aggregation is known with pairings and a long setup [HWW25]. These works supply the authentication compressors against which our proxy-validity boundary is defined.

Closest combined functionality. The closest application-level overlap is the pairing-based identity-based aggregate signcryption with re-encryption scheme of Choi, Jeong, and Byun [CJB26]. There, a UAV verifies and decrypts sensor reports before aggregate re-encryption to a base station. In CEA, the proxy remains plaintext-oblivious and the main question is which authentication facts

Table 1: Comparison with the principal lines of related work. "External" means that authentication is not integrated into the re-encryption transform. The final column isolates the security or trust boundary most relevant to this paper.

Work	Ciphertext transform	Authenticated object	Aggregation	Proxy sees plaintext	Security or trust boundary
BBS PRE [BBS98]	recipient change	external	none	no	Foundational atomic PRE; predates modern CCA and HRA notions.
Unidirectional PRE [AFGH06]	recipient change	external	none	no	Storage-oriented delegation; no aggregate authentication.
CCA PRE [CH07]	recipient change	external	none	no	Derivative-aware chosen-ciphertext security.
Unidirectional CCA PRE [LV08]	recipient change	external	none	no	Strong classical PRE baseline; pairing-based.
HRA PRE [Coh19]	honest reencryption	external	none	no	Corrupt recipient may receive an honest reencryption.
Tight HRA PRE [CCG+25, LCB+25]	recipient change	external	none	no	Tight or adaptive multi-challenge HRA security.
Joint signature and encryption [ADR02, FPS20]	none	ciphertext or committed message	none	—	Joint authenticity and confidentiality; no delegation.
Randomizable-ciphertext signatures [BF20]	rerandomize	plaintext equivalence class	none	no	Authentication adapts to rerandomization, not recipient change.
Pairing aggregates [BGLS03, HWW25]	none	signed messages	constant-size	—	Pairing-based aggregation in the ROM or plain model.
Schnorr half aggregation [CGKN21, CZ22]	none	Schnorr statements	one scalar plus commitments	—	Authentication compression independent of PRE validity.
Interactive DL aggregation [NRS26, ZWL26]	none	interactive or rebound transcripts	constant-size or domain-scaled	—	Requires interaction or certified transcript rebinding.
ID aggregate signcryption/PRE [CJB26]	UAV-mediated recipient change	sensor signcryptions	aggregate signcryption	yes	The UAV verifies and decrypts reports before re-encryption.
CEA (this work)	recipient change	invariant commitment	Schnorr half aggregation	no	Separates the G-CONF-CT attack from the candidate G-CONF-CLASS bound.

survive an algebraic transform. Table 1 locates that question across the adjacent PRE, signcryption, and aggregation literatures.

3 Model and notation

Let G be a cyclic additive group of prime order q, generated by P. All externally supplied points are validated, the identity is rejected where the algorithm requires a nonzero point, and scalars are canonically encoded in $[0, q)$. The implementation instantiates G with P-256 and SHA-256 with rejection sampling; the security analysis is stated in the random-oracle model.

A key reference is a tuple

$$\mathsf{KeyRef} = (\mathsf{id}, \mathsf{role}, \mathsf{version}, \mathsf{fingerprint}).$$

An origin class is the full origin decryption-key reference together with a tag τ and epoch e. Class keys are independent. A sender has Schnorr key x_S and public key $X_S = x_S P$. An origin class has decryption secret k and public key $K = kP$. A delegatee ℓ has secret k_ℓ and public key $K_\ell = k_\ell P$.

Registration accepts adversarial group elements under identities controlled by the adversary. Registration is nonextracting and omits proof of possession. The identity layer is assumed to prevent an adversary from registering a new key version for an honest identity. Corruptions are static in the confidentiality games considered here.

The oracles H_{d}, H_{c}, KDF, H_{mask}, and H_{Ω} are domain-separated. Hash inputs use typed, length-delimited encodings. The symbol λ denotes the bit length of message-opening digests and commitments; the implementation uses $\lambda = 256$.

Gap divisible computational Diffie–Hellman. The named assumption used below gives a solver an instance $(P, U = aP, V = bP)$, for independent nonzero a, b, and a decision oracle that returns one precisely when $Y = aX$ for a submitted pair (X, Y). The task is to output $(b/a)P$. The game adversary never receives the decision oracle. The output problem is the Divisible Computational Diffie–Hellman (DCDH) problem, shown equivalent to computational, inverse, and square CDH in the high-granularity sense [BDZ03]. The additional fixed-key decision oracle makes the exact interface a gap-DCDH variant; the artifact retains the label GDivDH for reproducibility.

4 The commit–encrypt–aggregate–delegate comparator

Figure 1 separates the three data lanes that matter for the security analysis. The aggregate authenticator binds the canonical statement containing c; the payload E is checked only after opening; and the proxy changes the unsigned point A. The last lane makes delegation possible and is also the source of the point-shifting attack.

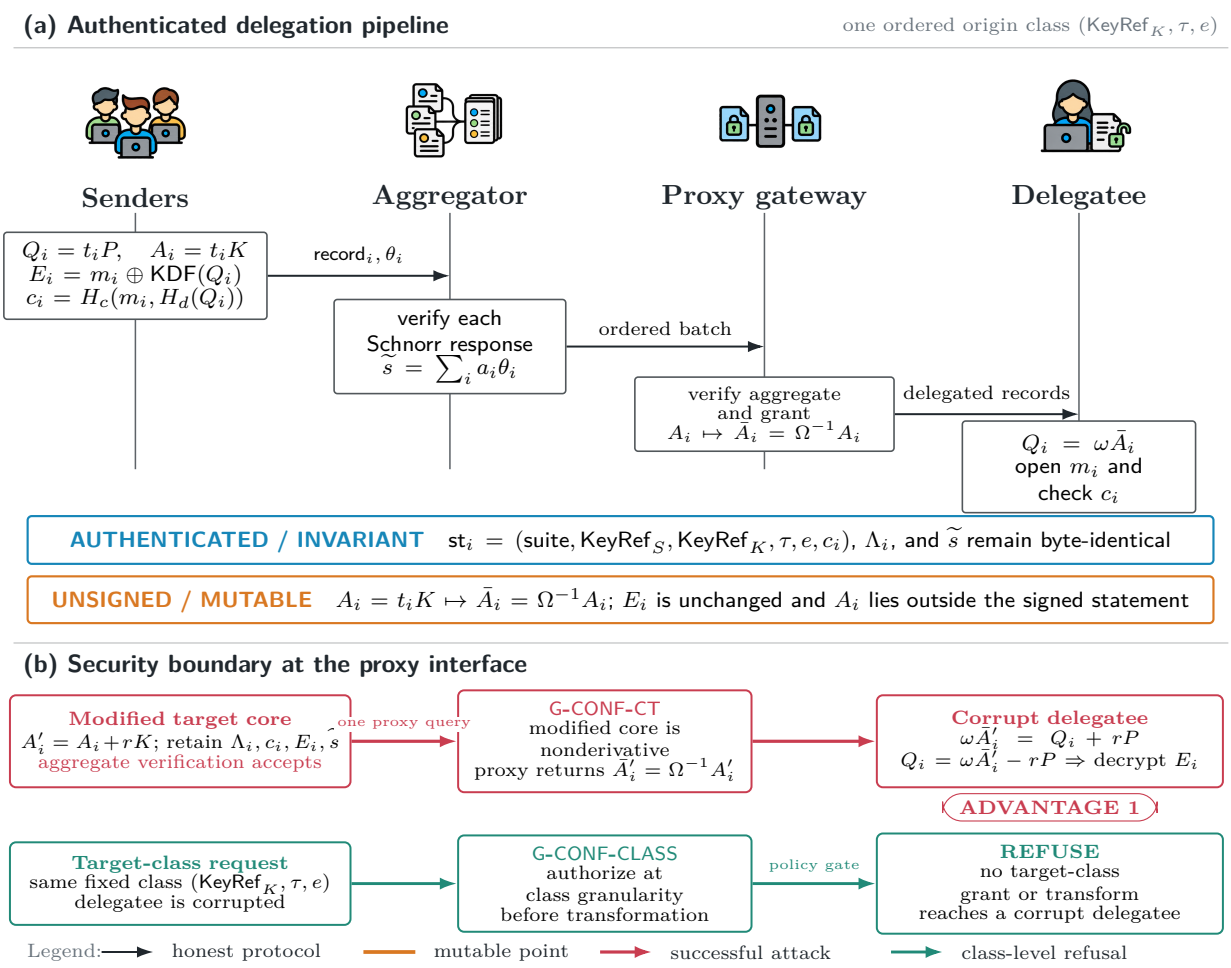


Figure 1: CipherAuth dataflow and its security boundary. Delegation preserves the commitment-bound aggregate statement exactly, but the proxy-validity decision depends on an unsigned, algebraically mutable point. The red path is the one-query attack against ciphertext-granular confidentiality; the weaker class-granular game refuses this service to corrupt target recipients.

4.1 Origin records

To encrypt a message m for class key $K = kP$, a sender samples $t \leftarrow \mathbb{Z}_q^*$ and computes

$$Q = tP, \qquad A = tK, \qquad d = H_{\mathsf{d}}(Q),$$
$$E = m \oplus \mathsf{KDF}(Q, |m|), \qquad c = H_{\mathsf{c}}(m, d).$$

The authenticated statement is the canonical encoding

$$\mathsf{st} = \mathsf{enc}(\mathsf{suite}, \mathsf{KeyRef}_S, \mathsf{KeyRef}_K, \tau, e, c).$$

The sender produces a Schnorr signature (Λ, θ) [Sch91] on st. The record body contains (Λ, c, A, E); the individual response θ is supplied to the aggregator but is omitted from the final compressed object.

For an ordered sequence, the aggregator verifies every individual signature and uses the Chen–Zhao prefix-coefficient form

$$a_1 = 1, \qquad a_i = H_2((\Lambda_1, X_1, \mathsf{st}_1), \ldots, (\Lambda_i, X_i, \mathsf{st}_i)) \quad (i \geq 2),$$

with nonzero coefficients where required by the transcript. It publishes the records and the scalar $\widetilde{s} = \sum_i a_i \theta_i$. Public verification recomputes the full ordered transcript and checks the corresponding aggregate Schnorr equation.

4.2 Grants and delegation

For a grant from class $K = kP$ to delegatee $K_\ell = k_\ell P$, the origin samples $\omega, t' \leftarrow \mathbb{Z}_q^*$ and computes

$$T = t'P, \qquad C = \mathsf{enc}(\omega) \oplus H_{\mathsf{mask}}(T, \mathsf{KeyRef}_K),$$
$$\Omega = \omega k \bmod q, \qquad \gamma = t'K_\ell, \qquad W = H_\Omega(\Omega).$$

The proxy retains the scalar Ω. The public token is (C, γ, W). The origin signs a grant binding the issuer, origin and delegatee key references, tag set, epoch, token digest, grant identifier, and suite.

After verifying the aggregate, grant, scope, and token digest, the proxy maps each record point as

$$A \longmapsto \bar{A} = \Omega^{-1} A.$$

It leaves (Λ, c, E), the aggregate scalar, and all authenticated sender statements unchanged. The delegatee recovers $T = k_\ell^{-1}\gamma$, unmasks ω, computes $Q = \omega\bar{A}$, and accepts the message only if $c = H_{\mathsf{c}}(m, H_{\mathsf{d}}(Q))$.

4.3 Exact commutation

Lemma 4.1 (Commutation identity)**.** *For every valid class grant and ordered list of valid origin submissions, the two operation orders produce the same records and the same aggregate response. Both open to the same ordered origin statements.*

Proof. The aggregate entry $(\Lambda_i, X_i, \mathsf{st}_i)$ contains the commitment c_i but not A_i. Delegation changes only A_i, so the coefficient sequence and $\widetilde{s}$ are identical in both orders. Correctness follows from

$$\omega\bar{A} = \omega(\omega k)^{-1}(tkP) = tP = Q.$$

□

This invariant explains why authentication survives recipient change and provides the algebraic baseline for the security separation that follows.

5 Two confidentiality granularities

Both games fix the target class and the honest/corrupted sets before setup. The challenge is an origin record for one of two equal-length messages. Challenge cores and proxy-produced derivatives are recorded. The decryption oracle refuses any aggregate containing a recorded core or derivative and replaces either challenge message with TEST, following a conservative RCCA-style rule.

In G-CONF-CLASS, the target class is never served to a corrupted recipient. GrantO may issue a target-class grant only to a statically honest delegatee, returns only the signed grant and public

token, and retains Ω at the honest proxy. ReEncO enforces the same admissibility rule. Proxy corruption is unavailable for the target class.

The stronger G-CONF-CT game permits ReEncO to serve a corrupted delegatee when the submitted core is not registered as the challenge or a derivative. This models a proxy that is expected to distinguish allowed and withheld ciphertexts within one class. Thus the games differ at the authorization boundary while retaining the same challenge distribution.

The scope game G-SCOPE extends G-CONF-CLASS by revealing complete tokens and allowing proxy–delegatee collusion for every non-target class of the same origin. The target class remains protected by the G-CONF-CLASS rules.

Figure 2 summarizes the separation. It also prevents a potential misreading of the positive result: G-CONF-CLASS is a coherent authorization game for this comparator and sits below HRA because its reencryption interface excludes corrupt target recipients.

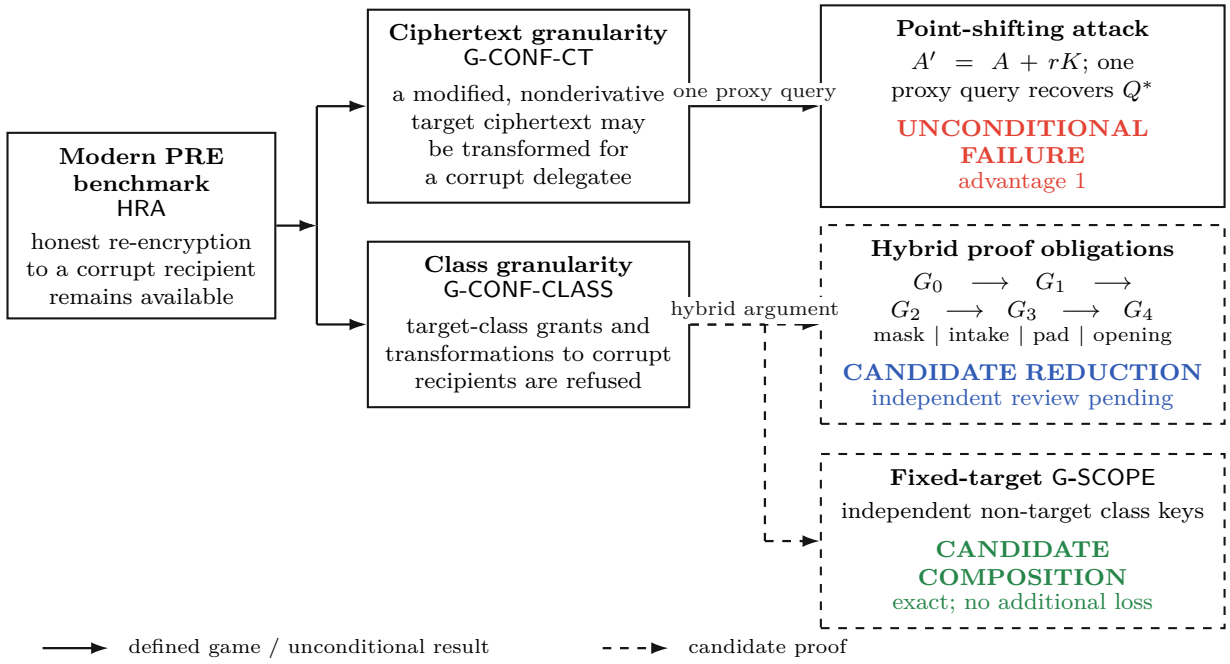


Figure 2: Game map, negative result, and positive proof obligations. The red branch is unconditional for the specified construction. The blue and green branches remain candidate reductions pending independent cryptographic review.

6 Per-ciphertext confidentiality is false

Theorem 6.1 (Point-shifting attack)**.** *The* CEA *comparator fails the per-ciphertext game* G-CONF-CT*. A corrupted delegatee wins with probability one, except for validation events that occur with negligible probability under honest challenge generation.*

Proof. Let the challenge contain $A^* = t^*K$, ciphertext component E^*, and authenticated commitment c^*. The adversary chooses known $r \in \mathbb{Z}_q^*$ and replaces the unsigned point by

$$A' = A^* + rK = (t^* + r)K.$$

It leaves Λ^*, c^*, E^* and the aggregate authentication unchanged. Public verification still accepts because the signed statement contains c^*, not A^*. The submitted core differs from the registered challenge core, so the G-CONF-CT proxy query is allowed.

For the corrupted delegatee's grant, the proxy returns

$$\bar{A}' = \Omega^{-1}A' = \omega^{-1}(Q^* + rP).$$

The delegatee uses its secret key and the public token to recover ω, then computes $Q' = \omega\bar{A}' = Q^* + rP$ and $Q^* = Q' - rP$. It decrypts E^* with $\mathsf{KDF}(Q^*)$, learning the challenge bit. The commitment check succeeds for the recovered challenge message. No decryption-oracle query is needed. □

The attack explains why authenticating an invariant commitment is insufficient for per-ciphertext proxy policy. A repair must give the proxy a sound way to validate the relation between A and the encryption randomness, such as a proof of knowledge tied to the complete ciphertext. Whether such proofs can be compressed while retaining per-record extraction is a separate construction and proof problem.

In-class collusion. The token also makes in-class key recovery explicit. The proxy knows $\Omega = \omega k$, while the delegatee recovers ω. Their coalition computes $k = \Omega\omega^{-1}$. Consequently, retained grant material cannot support revocation within a class and epoch. Independent future-epoch keys can limit future access, but they cannot retract access to an old class key.

7 Candidate class-granular confidentiality theorem

Define confidentiality advantage as $\left|\Pr[b' = b] - \frac{1}{2}\right|$. Let N_H be the number of statically honest delegatees, q_{pre} the number of distinct challenge-proof inputs queried before the challenge proof is installed, q_c the number of commitment-oracle queries, and Q_D the number of record-opening attempts in ordinary nonrefused decryption queries.

Theorem 7.1 (Candidate class-confidentiality bound)**.** *In the random oracle model, under GDivDH, the frozen* G-CONF-CLASS*-v1 interface satisfies*

$$\begin{aligned}\mathsf{Adv}_{\mathsf{G\text{-}CONF\text{-}CLASS}}(\mathcal{A}) \le \mathsf{Succ}_{\mathrm{GDivDH}}(\mathcal{B}_{\mathrm{chal}}) + N_H\mathsf{Succ}_{\mathrm{GDivDH}}(\mathcal{B}_{\mathrm{mask}}) \\ + \frac{q_{\mathrm{pre}}}{q-1} + \frac{q_c}{2^\lambda} + \frac{Q_D(q_c+1)}{2^\lambda},\end{aligned}$$

capped at one. The two GDivDH solvers are distinct reductions. The statement assumes static corruption and class-granular confidentiality; authenticity and concrete-hash analysis are treated separately.

7.1 Hybrid argument

Game G_0 is the real G-CONF-CLASS-v1 game. Target-class grants and proxy service to corrupted recipients are refused. Proxy outputs to honest delegatees are registered as derivatives, and decryption refuses registered cores and applies the TEST mask.

In G_1, for every honest target-class grant, replace the following value by uniform bytes:

$$C = \mathsf{enc}(\omega) \oplus H_{\mathsf{mask}}(T, \mathsf{KeyRef}_K).$$

Sample $\Omega \leftarrow \mathbb{Z}_q^*$ directly, set $W = H_\Omega(\Omega)$, and retain Ω at the proxy. Multiplication by fixed nonzero k is a bijection, so Ω has its real distribution. Until the adversary queries the exact context-bound mask point, the two games are identical. Section 7.2 reduces the bad event with loss N_H. In this hybrid every target-class proxy reply is exact for arbitrary input:

$$\bar{A} = \Omega^{-1} A.$$

This step is why no target-record knowledge extractor is needed.

Game G_2 installs a simulated challenge intake proof. Use the main GDivDH instance as $K^* = aP$ and $A^* = bP$. Sample a challenge and response (c_p, z_p), form $R^* = z_p K^* - c_p A^*$, reject the identity, and program the exact fresh proof input to c_p. Conditioned on freshness, the transcript has the honest Schnorr distribution. Each prequeried input fixes at most one encodable R^*, giving distance at most $q_{\mathrm{pre}}/(q-1)$.

In G_3, sample the challenge pad, digest d^*, and commitment c^* uniformly, and program $H_{\mathsf{c}}(m_b, d^*) = c^*$ if fresh. Until the adversary asks KDF or H_{d} at $Q^* = (b/a)P$, this has the real challenge distribution. The GDivDH decision oracle recognizes a point Q satisfying $A^* = aQ$, after which $\mathcal{B}_{\mathrm{chal}}$ outputs Q. A commitment prequery hits (m_b, d^*) with probability at most $q_c/2^\lambda$.

In G_4, ordinary decryption scans the recorded KDF and H_{d} point queries. The decision oracle recognizes a candidate Q satisfying $A = kQ$. If found, the simulator completes missing oracle values and verifies the opening. For a delegated record under an honest target grant, it first restores $A = \Omega\bar{A}$. If a real opening accepts without the correct point in either query list, the unseen $H_{\mathsf{d}}(Q)$ value must cause an earlier commitment collision or a fresh λ-bit guess. The probability is at most $(q_c+1)/2^\lambda$ per attempt, hence $Q_D(q_c+1)/2^\lambda$ in total.

Conditioned on none of these events, the challenge pad and commitment remain uniform at every allowed interface. The final view is independent of b. Summing the game differences gives Theorem 7.1.

7.2 Mask exposure reduction

The reduction guesses one of at most N_H honest delegatees and publishes its key as $U = aP$ from a GDivDH instance $(P, U, V = bP)$. For the j-th grant to that delegatee it samples nonzero r_j and publishes $\gamma_j = r_j V$. The hidden mask point is $T_j = r_j(b/a)P$. The reduction samples C_j and Ω_j uniformly, sets $W_j = H_\Omega(\Omega_j)$, signs the public token digest, and answers every proxy transformation as $\Omega_j^{-1} A$.

For each context-correct H_{mask} query X, the solver tests every stored grant with its decision oracle. It also tests earlier mask queries when a new grant is installed. If $\gamma_j = aX$, then $X = T_j$, and $r_j^{-1} X = (b/a)P$ solves the instance. No grant index or mask-query index is guessed; only the honest-delegatee guess contributes loss. Thus

$$\Pr[\mathsf{Bad}_{\mathrm{mask}}] \le N_H \mathsf{Succ}_{\mathrm{GDivDH}}(\mathcal{B}_{\mathrm{mask}}).$$

Static honesty is essential because later disclosure of k_ℓ would reveal $T_j = k_\ell^{-1}\gamma_j$ directly.

7.3 Why algebraic extraction is absent

An earlier route attempted to extract a numerical opening from every accepted compressed intake proof. That is unnecessary for G-CONF-CLASS. The only target-class proxy service is to honest delegatees, and the simulator knows the corresponding scalar Ω after the uniform-mask hybrid. It can therefore transform any group point, including the challenge and adversarial affine combinations,

without knowing an opening. Repeated-query and cross-grant linear relations are preserved because the simulated map is the real linear map.

For decryption, the GDivDH decision oracle identifies openings already present in the random-oracle query tables. A missing point query is accounted for by the explicit commitment-guessing term. The aggregate intake equation remains part of the implemented proxy interface; Theorem 7.1 instead relies on exact scalar simulation and decision-oracle recognition of queried openings.

8 Out-of-class scope

In G-SCOPE-v1, the adversary may receive complete grants and collude with the proxy and delegatees for every class of the same origin except the phase-zero target. Each class is identified by the full origin key reference, tag, and epoch.

Theorem 8.1 (Candidate exact scope composition)**.** *For a target class fixed before key generation,*

$$\mathsf{Adv}_{\mathsf{G\text{-}SCOPE}}(\mathcal{A}) \le \mathsf{Adv}_{\mathsf{G\text{-}CONF\text{-}CLASS}}(\mathcal{B}).$$

The composition adds no assumption or additive loss. Substituting Theorem 7.1 gives the same right-hand side.

Proof. The reduction forwards the fixed target class to its class-confidentiality challenger and generates every other class key independently. It knows those scalars and answers non-target grants, token disclosure, proxy corruption, re-encryption, and decryption exactly, including recovery of a granted non-target class key by a proxy–delegatee coalition.

Cross-class substitution is rejected by the canonical bindings: sender statements bind the complete origin key reference, tag, and epoch; grants and token digests bind the origin and delegatee references and public token fields; the proxy checks the grant against the aggregate context. The mask-oracle input also includes the full origin class reference. Global KDF and H_{d} queries are already handled by the main class solver. The simulated view is therefore exact. □

An adaptive target selected after observing N eligible class keys incurs at least the elementary random-index factor N under this reduction.

9 Authentication, costs, and artifact evidence

Aggregate authentication follows through a reduction in which a fresh honest-sender statement in an accepted aggregate yields either a chosen-key IASchnorr forgery or a collision in H_{c}. Grant authenticity reduces to Schnorr EUF-CMA under identity-authenticated key registration. The proof ledger tracks the exact Chen–Zhao loss transcription for the aggregate theorem.

The implemented origin record adds one compressed group point Λ and one 32-byte commitment c to the hashed-ElGamal core, plus one 32-byte aggregate response per batch. The public token contains 97 bytes before framing: (C, γ, W). In the canonical prototype wire format, an eight-record batch with 32-byte messages is 1337 bytes at the origin and 1665 bytes after delegation; the signed grant and public token add 328 bytes.

The artifact contains 286 unique tests: 270 for the current chain and 16 preserved historical diagnostics. It also replays 33 deterministic traces and compiles 266 Python files. The integrated path is

$$\text{sender} \to \text{compressed intake} \to \text{proxy} \to \text{canonical wire} \to \text{delegatee opening}.$$

All tests pass with zero failures, errors, or skips. The source manifest is checked separately so review material cannot be inserted into the frozen codebase without detection.

The complete review artifact, including source, tests, and deterministic traces, is available at `https://anonymous.4open.science/r/CipherAuth-7A03/README.md`.

The test suite covers exact commutation, origin/delegatee opening equality, canonical serialization, malformed encodings, scope bindings, the A-mauling attack, the in-class key-recovery non-goal, token-hybrid algebra, challenge-proof programming, and the quantitative loss calculator. These tests validate the executable and algebraic claims, while the reductions carry the computational claims. The machine-readable summary records these evidence classes separately.

10 Discussion

The construction exposes a useful distinction between authentication invariance and proxy-validity. Signing an invariant commitment is sufficient for exact aggregate commutation and end-recipient integrity. It is insufficient when a proxy must enforce policy at ciphertext granularity, because the proxy cannot open the commitment. A proxy-checkable validity proof could close that gap, but then knowledge soundness and its interaction with aggregation become part of the main security argument.

Class granularity is coherent only when the deployment accepts its consequences. A class is an authorization unit rather than metadata alone. A target class cannot be served to corrupted recipients, and collusion on a granted class reveals that class key. Short epochs and independent keys limit future exposure but do not provide cryptographic revocation of retained material.

The reductions use the GDivDH decision oracle to detect the challenge opening and token-mask exposure while retaining the standard DCDH computational target. Twin-DH techniques offer a concrete route for eliminating decision oracles in related settings [CKS08]; adapting that route to the present interface is the natural next theoretical step.

11 Conclusion

Commitment signing makes aggregation and delegation commute, but the same design leaves the encryption point unchecked at the proxy. The resulting gap is sharp: per-ciphertext confidentiality fails by a one-query mauling attack, while a class-granular notion—strictly weaker than HRA because corrupt target recipients are refused—admits an exact token simulation and an explicit candidate reduction. Independent class keys extend that result to out-of-class collusion without further loss. Together, the attack, game separation, candidate quantitative bound, and exact scope composition define the achieved security frontier of the comparator.

A Frozen oracle rules

For clarity, this appendix records the target-class rules used by the candidate theorem. GrantO refuses corrupted target-class delegatees. For an honest delegatee it returns only the signed grant and public (C, γ, W), retaining Ω at the proxy. ReEncO requires a recorded admissible grant and public/intake acceptance, refuses corrupted target-class delegatees for every input, and registers outputs derived from a challenge core. DecO first refuses every registered core or derivative; otherwise it performs ordinary opening and masks either challenge message with TEST. Target-proxy corruption is unavailable.

The challenge registers $(\mathsf{KeyRef}_K, A^*, E^*)$. Re-encryption of the exact challenge to an honest delegatee is allowed and the result becomes a derivative. This query is simulatable as $\Omega^{-1}A^*$; no opening of A^* is needed.

B Proof-obligation ledger

Table 1. Status at the release date.

Item	Status
Exact aggregate/delegate commutation	Algebraic identity; tested
Per-ciphertext A-mauling attack	Concrete attack; tested
In-class proxy–delegatee key recovery	Algebraic attack; tested
G-CONF-CLASS-v1 quantitative bound	Candidate proof; independent review required
G-SCOPE-v1 fixed-target composition	Candidate exact composition; independent review required
Aggregate sender authenticity	Known-theorem composition sketch; exact loss transcription and review required
Grant authenticity	Schnorr reduction sketch; review required
Concrete SHA-256 security	Not claimed
Machine-checked cryptographic proof	None

References

[ADR02] Jee Hea An, Yevgeniy Dodis, and Tal Rabin. On the security of joint signature and encryption. In *Advances in Cryptology – EUROCRYPT 2002*, volume 2332 of *Lecture Notes in Computer Science*, pages 83–107, 2002. Full version: IACR ePrint 2002/046.

[AFGH06] Giuseppe Ateniese, Kevin Fu, Matthew Green, and Susan Hohenberger. Improved proxy re-encryption schemes with applications to secure distributed storage. *ACM Transactions on Information and System Security*, 9(1):1–30, 2006.

[BBS98] Matt Blaze, Gerrit Bleumer, and Martin Strauss. Divertible protocols and atomic proxy cryptography. In *Advances in Cryptology – EUROCRYPT '98*, volume 1403 of *Lecture Notes in Computer Science*, pages 127–144, 1998.

[BDZ03] Feng Bao, Robert H. Deng, and Huafei Zhu. Variations of diffie–hellman problem. In *Information and Communications Security – ICICS 2003*, volume 2836 of *Lecture Notes in Computer Science*, pages 301–312, 2003.

[BF20] Balthazar Bauer and Georg Fuchsbauer. Efficient signatures on randomizable ciphertexts. In *Security and Cryptography for Networks – SCN 2020*, volume 12238 of *Lecture Notes in Computer Science*, pages 359–381, 2020.

[BGLS03] Dan Boneh, Craig Gentry, Ben Lynn, and Hovav Shacham. Aggregate and verifiably encrypted signatures from bilinear maps. In *Advances in Cryptology – EUROCRYPT 2003*, volume 2656 of *Lecture Notes in Computer Science*, pages 416–432, 2003.

[CCG+25] Aloni Cohen, David Bruce Cousins, Nicholas Genise, Erik Kline, Yuriy Polyakov, and Saraswathy R. V. HRA-secure homomorphic lattice-based proxy re-encryption with tight security. *IACR Communications in Cryptology*, 2(1), 2025.

[CGKN21] Konstantinos Chalkias, Francois Garillot, Yashvanth Kondi, and Valeria Nikolaenko. Non-interactive half-aggregation of eddsa and variants of schnorr signatures. IACR Cryptology ePrint Archive, Report 2021/350, 2021.

[CH07] Ran Canetti and Susan Hohenberger. Chosen-ciphertext secure proxy re-encryption. In *Proceedings of the 14th ACM Conference on Computer and Communications Security*, pages 185–194, 2007. Full version: IACR ePrint 2007/171.

[CJB26] Jae Hyun Choi, Ik Rae Jeong, and Jin Wook Byun. Id-based aggregate signcryption with re-encryption for secure uav networks. *Information Sciences*, 754:123676, 2026.

[CKN03] Ran Canetti, Hugo Krawczyk, and Jesper Buus Nielsen. Relaxing chosen-ciphertext security. In *Advances in Cryptology – CRYPTO 2003*, volume 2729 of *Lecture Notes in Computer Science*, pages 565–582, 2003. Full version: IACR ePrint 2003/174.

[CKS08] David Cash, Eike Kiltz, and Victor Shoup. The twin diffie–hellman problem and applications. In *Advances in Cryptology – EUROCRYPT 2008*, volume 4965 of *Lecture Notes in Computer Science*, pages 127–145, 2008.

[Coh19] Aloni Cohen. What about bob? the inadequacy of CPA security for proxy reencryption. In *Public-Key Cryptography – PKC 2019, Part II*, volume 11443 of *Lecture Notes in Computer Science*, pages 287–316, 2019.

[CZ22] Yanbo Chen and Yunlei Zhao. Half-aggregation of schnorr signatures with tight reductions. In *Computer Security – ESORICS 2022, Part II*, volume 13555 of *Lecture Notes in Computer Science*, pages 385–404, 2022. Full version: IACR ePrint 2022/222.

[FPS20] Georg Fuchsbauer, Antoine Plouviez, and Yannick Seurin. Blind schnorr signatures and signed elgamal encryption in the algebraic group model. In *Advances in Cryptology – EUROCRYPT 2020*, Lecture Notes in Computer Science, 2020. Full version: IACR ePrint 2019/877.

[HWW25] Susan Hohenberger, Brent Waters, and David J. Wu. Pairing-based aggregate signatures without random oracles. In *Advances in Cryptology – ASIACRYPT 2025, Part VI*, volume 16250 of *Lecture Notes in Computer Science*, pages 271–303, 2025.

[LCB+25] Yunhao Ling, Jie Chen, Zijian Bao, Man Ho Au, Luping Wang, and Haifeng Qian. Tightly, adaptively secure proxy re-encryption in multi-challenge setting. In *Advances in Cryptology – ASIACRYPT 2025, Part VI*, volume 16250 of *Lecture Notes in Computer Science*, pages 174–204, 2025.

[LV08] Benoît Libert and Damien Vergnaud. Unidirectional chosen-ciphertext secure proxy re-encryption. In *Public Key Cryptography – PKC 2008*, volume 4939 of *Lecture Notes in Computer Science*, pages 360–379, 2008.

[NRS26] Jonas Nick, Tim Ruffing, and Yannick Seurin. DahLIAS: Discrete logarithm-based interactive aggregate signatures. In *Advances in Cryptology – EUROCRYPT 2026, Part I*, Lecture Notes in Computer Science, pages 215–245, 2026.

[Sch91] Claus-Peter Schnorr. Efficient signature generation by smart cards. *Journal of Cryptology*, 4(3):161–174, 1991.

[Zhe97] Yuliang Zheng. Digital signcryption or how to achieve cost(signature & encryption) $\ll$ cost(signature) + cost(encryption). In *Advances in Cryptology – CRYPTO '97*, volume 1294 of *Lecture Notes in Computer Science*, pages 165–179, 1997.

[ZWL26] Lingqing Zhao, Liping Wang, and Yu Li. Composable aggregation of schnorr signatures via transcript rebinding. In *Information and Communications Security – ICICS 2026, Part I*, volume 17214 of *Lecture Notes in Computer Science*, pages 282–302, 2026.